AYUSH GUPTA

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PROFESSIONAL SUMMARY

Machine Learning Engineer with experience in building end-to-end ML systems, including predictive modeling, computer vision pipelines, and LLM-based applications. Skilled in data preprocessing, model optimization, and deploying scalable ML solutions using Python and modern ML frameworks.

WORK EXPERIENCE

LPU Innovation Studio *Jalandhar, Punjab | Jul 2023 – Aug 2023*

R&D Intern

- Participated in innovation and intellectual property training, gaining exposure to patent filing processes and IP fundamentals
- Conducted research-driven projects focusing on Machine Learning applications in Emotion Detection, including literature review and comparative analysis of ML approaches
- Collaborated in a cross-functional team to design and implement a forge counter mechanism, contributing to improved production accuracy and operational efficiency
- Applied analytical thinking and prototyping methodologies to bridge theoretical research concepts with practical implementation

AllSoft Solutions Ltd. (An IBM Partner) *Mohali, Punjab | Aug 2023 – Jul 2023*

Machine Learning Intern

- Built an end-to-end churn prediction model for the E-commerce/DTH domain, addressing account-level churn impacting multi-customer revenue
- Performed data preprocessing, feature engineering, and EDA to identify key churn drivers (service score, tenure, customer care interactions, revenue trends)
- Handled imbalanced data using SMOTE-Tomek and trained Logistic Regression, Decision Tree, Random Forest, and Neural Network (MLP) models using scikit-learn
- Optimized models with GridSearchCV, achieving ~88% test accuracy with the Neural Network while maintaining strong precision, recall, F1-score, and ROC-AUC
- Proposed a revenue-safe, segmented retention strategy targeting high-risk accounts without blanket discounts, aligned with Revenue Assurance constraints
- Designed complete ML pipeline including preprocessing, training, and evaluation

PROJECTS

RAG Optimizer *Apr 2026*

- Implemented an evaluation framework using RAGAS to measure faithfulness and answer relevance, enabling iterative improvements that boosted query response accuracy by 25%.
- Developed a RAG evaluation framework using RAGAS to measure faithfulness and answer relevance
- Designed reward-based optimization strategy to improve retrieval quality
- Achieved ~25% improvement in response accuracy through iterative tuning

Persona Adaptive Customer Support Agent *Oct 2025*

- Engineered an AI-driven support agent that dynamically adjusts conversational tone and resolution strategies based on real-time customer persona analysis and historical interaction data.
- Built a persona-adaptive support system using LLMs and RAG
- Implemented NLP-based persona classification to adjust response tone dynamically
- Integrated contextual memory and prompt engineering for improved response relevance
- Designed backend architecture supporting real-time interaction

SKILLS

Languages: Python, C++, SQL
Machine Learning: Scikit-learn, TensorFlow, PyTorch
Computer Vision: OpenCV, CNNs
NLP/LLM: Transformers, RAG, Prompt Engineering
Backend: FastAPI, REST APIs
Data: Pandas, NumPy

EDUCATION

Lovely Professional University *Jalandhar, Punjab|2025*
Computer Science and Engineering

Ursuline Convent Sr. Sec. School *Greater Noida, Uttar Pradesh|2020*
12th with PCMB

Ursuline Convent Sr. Sec. School *Greater Noida, Uttar Pradesh|2018*
10th with Science

CERTIFICATIONS

Neural Networks and Deep Learning, DeepLearning.AI *Jan 2026*
Supervised Machine Learning: Regression and Classification, DeepLearning.AI *Oct 2025*

Foundations of Data Science, Google *Apr 2025*
Natural Language Processing with Classification and Vector Spaces, DeepLearning.AI *Feb 2025*
Interactive Computer Graphics, University of Tokyo *Feb 2025*
Introduction to Shader Graphics, Coursera *Feb 2025*
Begin with Python, Programming Pathshala *Jun 2023*
Divide and Conquer, Sorting and Searching, and Randomized Algorithms, Coursera *Dec 2022*
Fundamentals of Deep Learning, NVIDIA *May 2022*

VOLUNTEERING & LEADERSHIP

National Service Scheme (LPU-NSS NAVJEEVAN)
Jalandhar | Sept 2023 - Present

Proud Batchmate

- Led a team of 30 volunteers in community outreach, conducting blood donation camps and teaching computer literacy to students in local schools.
- Executed social impact initiatives serving 100+ residents and participated in intensive winter service camps focused on community wellness and facility improvement.
- Participated in intensive winter service camps, focusing on organizational facility improvement, community service, and wellness initiatives.

Hostel Directing Committee
Jalandhar | Aug 2023 - Aug 2024

Treasurer and Hostel Manager

- Coordinated hostel operations and facility management, improving living standards for 500+ residents by streamlining communication between student committees and administration.
- Managed operations and budget allocation for student-led initiatives, ensuring transparency and accountability in fund distribution across diverse hostel events.

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING (SCSE)
Jalandhar | Aug 2022 - Mar 2023

Team Coordinator

- Coordinated 4 university-wide contests (2 cultural, 2 technical), managing outreach for 2,000+ students and providing technical support for 700+ live participants.
- Led team efforts to secure 1st place in 2 competitive events, demonstrating strong leadership and collaborative problem-solving.

LANGUAGES

Hindi (Native), English (Professional)